

$$9) \ y = -(-2x + 6)^{\frac{1}{2}}; \ [-2, 3]$$

$$10) \ y = -(-5x + 25)^{\frac{1}{2}}; \ [3, 5]$$

For each problem, determine if the Mean Value Theorem can be applied. If it can, find all values of c that satisfy the theorem. If it cannot, explain why not.

$$11) \ y = -\frac{x^2}{4x + 8}; \ [-3, -1]$$

$$12) \ y = \frac{-x^2 + 9}{4x}; \ [1, 3]$$

$$13) \ y = -(6x + 24)^{\frac{2}{3}}; \ [-4, -1]$$

$$14) \ y = (x - 3)^{\frac{2}{3}}; \ [1, 4]$$

Critical thinking question:

15) Use the Mean Value Theorem to prove that $|\sin a - \sin b| \leq |a - b|$ for all real values of a and b where $a \neq b$.